Unit Accel

SKU:U056













Description

Unit Accel is a motion data sensor with a built-in **ADXL345** three-axis accelerometer, capable of measuring acceleration up to **±16g** (13-bit resolution). The output data format is 16-bit two' s complement and can be accessed via the I2C digital interface (address 0x53). An accelerometer is a device that measures acceleration forces. These forces can be static (such as gravity) or dynamic, caused by movement or vibration of the accelerometer. By measuring acceleration due to gravity, you can calculate the tilt angle of the device relative to the horizontal plane. By analyzing dynamic acceleration, you can determine how the device is moving.

Features

- Low power consumption design
- Single-tap / double-tap detection
- Activity / inactivity monitoring
- Free-fall detection
- I2C digital interface

Includes

- 1 x Unit Accel
- 1 x HY2.0-4P Grove Cable (20cm)

Applications

- Building and structural monitoring
- Navigation
- Orientation sensing

Specifications

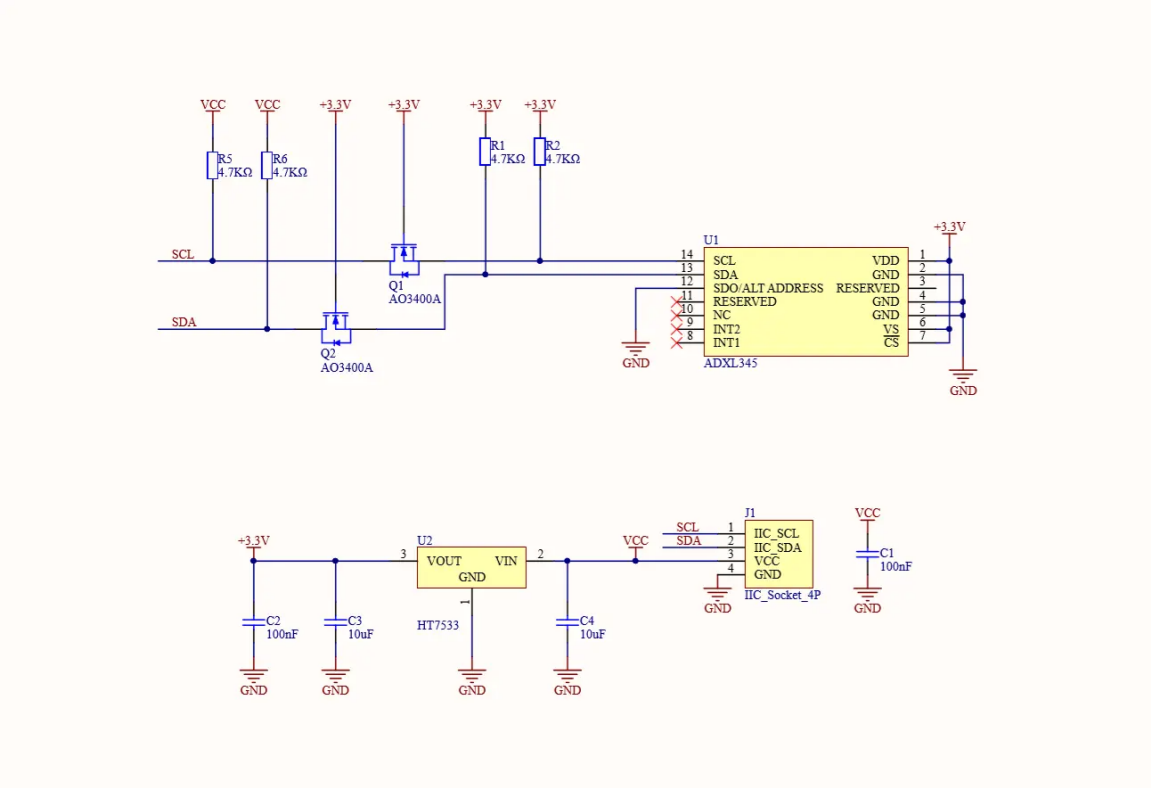
Specification	Parameter
Measurement Range	±16g
Communication Interface	I2C Communication @ 0x53
Resolution	10-bit fixed resolution, resolution increases with g range, up to 13-bit at ±16g
Power Consumption	VS = 2.5 V (typical), as low as 23 µA in measurement mode, 0.1 µA in standby mode
I/O Voltage Range	1.7V ~ Vs
Operating Temperature	−40°C ~ +85°C
Product Size	32.0 x 24.0 x 8.0mm
Product Weight	4.7g
Package Size	138.0 x 93.0 x 9.0mm
Gross Weight	9.4g

PinMap

Unit Accel

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

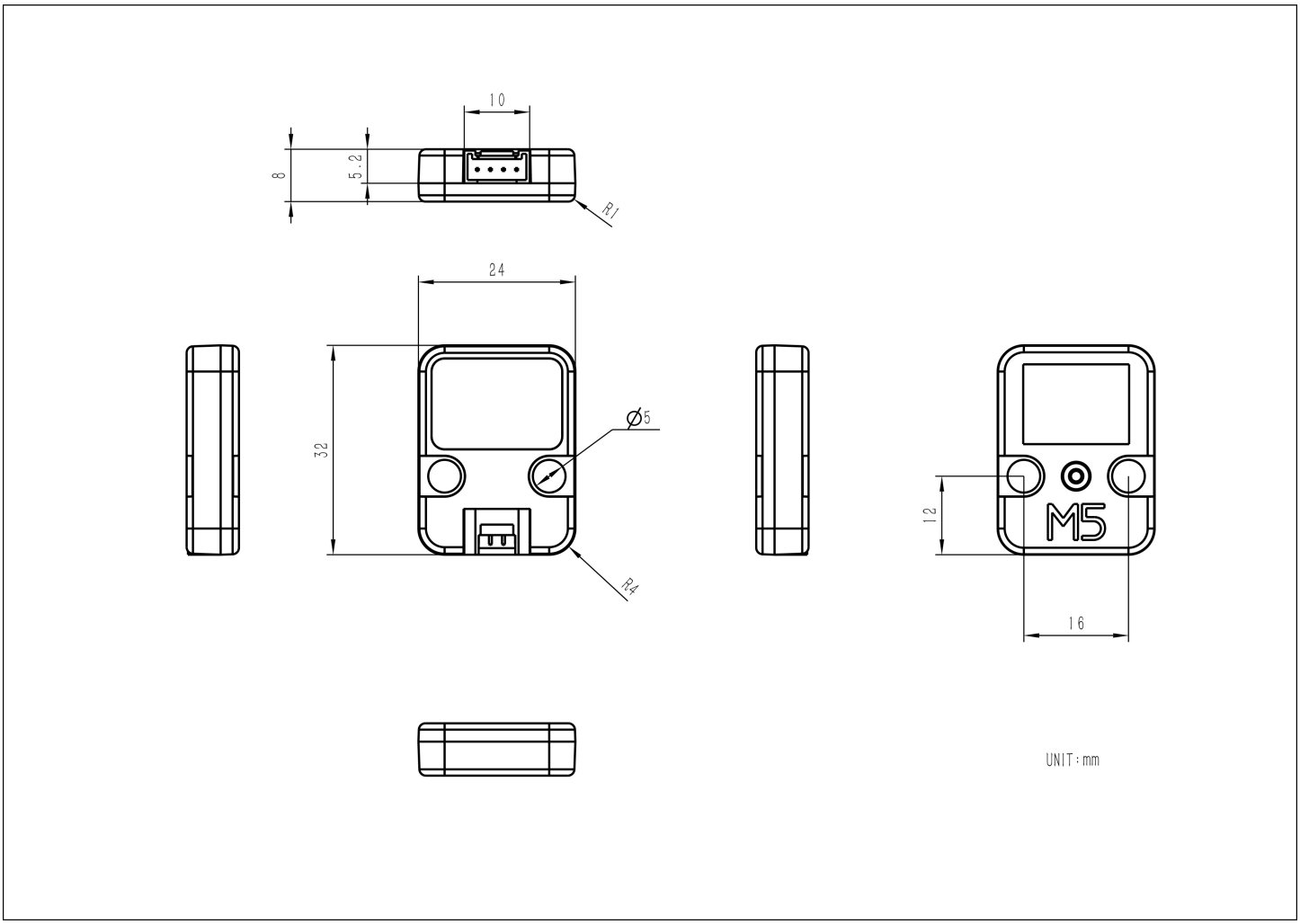
Schematics



Datasheets

- ADXL345

Model Size



Structure

- Unit Accel Structure Files

Softwares

Arduino

- Unit Accel Arduino Driver Library
- Unit Accel Example with M5Core
- Unit Accel Example with M5StickC
- Unit Accel Example with M5Core2

UiFlow1

- Unit Accel UiFlow1 Docs

UiFlow2

- Unit Accel UiFlow2 Docs

Easyloader

Easyloader	Download	Note
Unit Accel example with M5Core	download	/

Video

ACCEL.mp4